The iterated fiber sequence: fibers of the successive maps are the loop spaces of X and Y

The Univalent Foundations Program

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Editorial note. The counted statement is the author's 'thm:fiber-of-the-fiber' (source lines 1001-1008, Lemma 8.4.4); the proof environment starts on the next line, so the pairing is by adjacency.

Editorial note. The retained context is the author's exercise characterising pointed equivalences, which the proof invokes by label when checking that the equivalence is pointed.

Editorial note. The retained counter values (chapter 8, section 4, theorem offset 3) make the counted lemma print as Lemma 8.4.4, matching the original book numbering.

Editorial note. The single external reference is the associativity result for sigma types used at source line 1014; its number is taken from the label-number dump of the pinned book snapshot.

Editorial scope. One original result of the HoTT Book, counted once: the iterated fiber sequence (Lemma 8.4.4).

Allowed standard prerequisites (editorial).

- Fiber of a pointed map. The fiber of f : X -> Y at y_0 is Sigma x:X (f x = y_0), pointed at (x_0, f_0).. It is the type whose iterated fibers the statement describes.
- Iterated maps f^(n) of the fiber sequence. The maps f^(n) are the successive projections and connecting maps of the fiber sequence, with f^(n-1) o f^(n) = 0.. The three parts of the statement are about f^(1), f^(2) and f^(3).
- Associativity of sigma types. Sigma types reassociate up to equivalence.. It is the first step of the displayed chain at source line 1014.

Source reference. Original exercise 2.10: sources/hott-homotopy/book-labelnumbers-first-edition.tex, source lines 1402–1402. Original result cited by the retained text but located outside the retained span. It is a navigation pointer: the counted statement's own author proof is retained line for line, and no part of it is replaced by this reference. The reference number is taken from the label-number dump of the pinned book snapshot.

Source reference. Original section 4: sources/hott-homotopy/book-labelnumbers-first-edition.tex, source lines 164–164. Pointer to the chapter where pointed

equivalences and their coherent formulations are developed; the retained exercise invokes it by label. It is a navigation pointer only: the counted lemma's own author proof is retained line for line and nothing is replaced by this reference. The reference number is taken from the label-number dump of the pinned book snapshot.

Exercise 8.1. Define a **pointed equivalence** to be a pointed map whose underlying function is an equivalence.

(i) Show that the type of pointed equivalences between pointed types (X, x_0) and (Y, y_0) is equivalent to $(X, x_0) =_{\mathcal{U}_\bullet} (Y, y_0)$.
(ii) Reformulate the notion of pointed equivalence in terms of a pointed quasi-inverse and pointed homotopies, in one of the coherent styles from original section 4.

Lemma 8.4.4. *Let $f : X \to Y$ be a pointed map of pointed spaces. Then:*

(i) The fiber of $f^{(1)} :\equiv \mathsf{pr}_1 : \mathsf{fib}_f(y_0) \to X$ is equivalent to ΩY.
(ii) Similarly, the fiber of $f^{(2)} : \Omega Y \to \mathsf{fib}_f(y_0)$ is equivalent to ΩX.
(iii) Under these equivalences, the pointed map $f^{(3)} : \Omega X \to \Omega Y$ is identified with the pointed map $\Omega f \circ (-)^{-1}$.

Proof. For (i), we have

$$\begin{aligned}
\mathsf{fib}_{f^{(1)}}(x_0) &:\equiv \sum_{z:\mathsf{fib}_f(y_0)} (\mathsf{pr}_1(z) = x_0) \\
&\simeq \sum_{(x:X)} \sum_{(p:f(x)=y_0)} (x = x_0) && \text{(by original exercise 2.10)} \\
&\simeq (f(x_0) = y_0) && (\text{as } \textstyle\sum_{(x:X)}(x = x_0) \text{ is contractible}) \\
&\simeq (y_0 = y_0) && \text{(by } (f_0 \cdot -)) \\
&\equiv \Omega Y.
\end{aligned}$$

Tracing through, we see that this equivalence sends $((x, p), q)$ to ${f_0}^{-1} \cdot f\big(q^{-1}\big) \cdot p$, while its inverse sends $r : y_0 = y_0$ to $((x_0, f_0 \cdot r), \mathsf{refl}_{x_0})$. In particular, the basepoint $((x_0, f_0), \mathsf{refl}_{x_0})$ of $\mathsf{fib}_{f^{(1)}}(x_0)$ is sent to ${f_0}^{-1} \cdot f\big({\mathsf{refl}_{x_0}}^{-1}\big) \cdot f_0$, which equals refl_{y_0}. Hence this equivalence is a pointed map (see Exercise 8.1). Moreover, under this equivalence, $f^{(2)}$ is identified with $\lambda r.\, (x_0, f_0 \cdot r) : \Omega Y \to \mathsf{fib}_f(y_0)$.

item (ii) follows immediately by applying (i) to $f^{(1)}$ in place of f. Since $(f^{(1)})_0 :\equiv \mathsf{refl}_{x_0}$, under this equivalence $f^{(3)}$ is identified with the map $\Omega X \to \mathsf{fib}_{f^{(1)}}(x_0)$ defined by $s \mapsto ((x_0, f_0), s)$. Thus, when we compose with the previous equivalence $\mathsf{fib}_{f^{(1)}}(x_0) \simeq \Omega Y$, we see that s maps to ${f_0}^{-1} \cdot f\big(s^{-1}\big) \cdot f_0$, which is by definition $(\Omega f)(s^{-1})$. We omit the proof that this is an equality of pointed maps rather than just of functions. □